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Effect of argon addition on plasma parameters and dust charging in hydrogen plasma

B. Kakati,[1,a)] S. S. Kausik,[1] M. Bandyopadhyay,[2] B. K. Saikia,[1] and Y. C. Saxena[2]
[1]*Centre of Plasma Physics-Institute for Plasma Research, Nazirakhat, Sonapur, Kamrup 782402, Assam, India*
[2]*Institute for Plasma Research, Bhat, Gandhinagar 382 428, India*

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Experimental results on effect of adding argon gas to hydrogen plasma in a multi-cusp dusty plasma device are reported. Addition of argon modifies plasma density, electron temperature, degree of hydrogen dissociation, dust current as well as dust charge. From the dust charging profile, it is observed that the dust current and dust charge decrease significantly up to 40% addition of argon flow rate in hydrogen plasma. But beyond 40% of argon flow rate, the changes in dust current and dust charge are insignificant. Results show that the addition of argon to hydrogen plasma in a dusty plasma device can be used as a tool to control the dust charging in a low pressure dusty plasma. © *2014 AIP Publishing LLC*. [http://dx.doi.org/10.1063/1.4898858]

I. INTRODUCTION

Low-pressure hydrogen plasmas and hydrogen containing plasmas play an important role in variety of environments including planetary ionospheres, interstellar media,[1] and different industrial process[2–8] such as thin film deposition, fabrication of solar cell,[2] surface cleaning and passivation techniques to grow high quality diamond films,[3] and in controlled fusion devices.[8–12] Plasma with low electron temperature ~1 eV is very essential for effective production of negative ions, to reduce the unwanted chemical reactions, excitation of unwanted atomic and molecular states.

Different researchers have studied the effect of noble gas addition into the hydrogen plasma discharge.[10–15] It is observed by different researchers that addition of argon into hydrogen plasma enhances the negative hydrogen ion density upto an optimum partial pressure of argon.[10–12] The main process which contributes to the formation of negative hydrogen ion in low pressure hydrogen plasma is the dissociative attachment (DA) of low energetic electron (0.75 eV) with a rovibrationally excited hydrogen molecule[10–12]

$$e + H_2(v^{//}) \rightarrow H^- + H. \quad (1)$$

The higher rovibrational states are populated by radiative decay from singlet electronic states excited by collisions of ground-state molecules with energetic primary electrons. For an efficient use of the highly rovibrationally excited molecules, the electron density should be as high as possible. Curran *et al.*[10] suggested that the resonant energy exchange between excited argon atoms and hydrogen molecules can be the contributing factor for enhancing the excited H_2 density in argon mixed hydrogen plasma, as the argon has excited states with energies very close to those of the vibrational states of the first electronic excited state $(B^1\sum_u^+)$ of the hydrogen molecule. An increase in excited H_2 molecule density due to the addition of argon enhances the negative hydrogen ion formation in such plasma.

Bacal *et al.*[12] reported that adding argon to hydrogen plasma with low base pressure enhances the density of low-energy electrons but does not increase the formation of rovibrationally excited hydrogen molecules. They concluded that the increase of the negative-ion density by adding argon into a low-pressure hydrogen discharge is probably due to the increase of the low-energy electron density.

Our present study is on dust charging with an objective to control it by introducing different percentage of argon in hydrogen plasma. The addition of argon in hydrogen plasma affects on electron density, ion density, electron temperature, effective ion mass, negative hydrogen ion density, and degree of dissociation. The charge accumulated on dust grains is studied in such plasma for different percentage of argon flow rates to identify the role of electron density, ion density, electron temperature, effective ion mass, and degree of dissociation on it.

There are different types of experimental techniques used to measure dust charge in plasma background.[16–23] In the present experiment, the charge accumulated on a single dust grain is calculated using capacitance model and the current carried by a single dust is measured using a sensitive Faraday cup.[24,25] The present work is a part of an integrated study being carried out to identify different external controlling parameters,[26,27] which influences the dust charging in plasma background.

II. EXPERIMENTAL SET UP AND METHODOLOGY

The experimental setup consists of two stainless steel chambers placed one above other as shown in Fig. 1. The lower chamber is the plasma chamber that is placed horizontally and vertical upper chamber facilitates the holding of dust dropper. A steady state low pressure, direct current plasma is produced by hot cathode filament discharge technique. The plasma discharge chamber is made of a cylindrical stainless steel chamber of 100 cm in length and 30 cm in diameter. The dust unit consists of a cylindrical stainless

[a)]Electronic mail: bharatkakati15@gmail.com

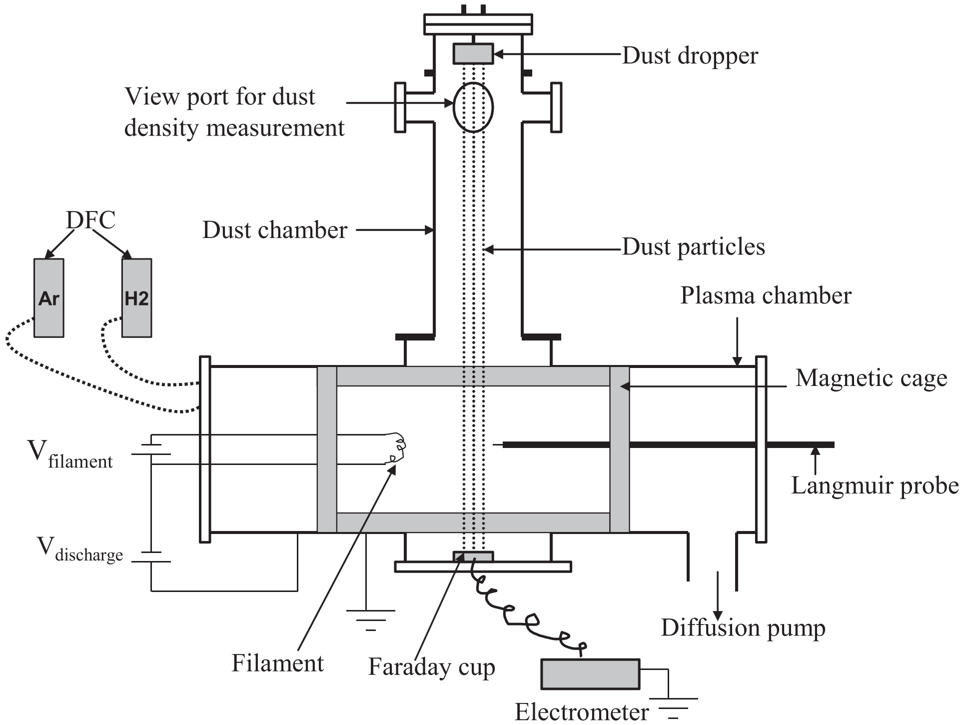

FIG. 1. Schematic diagram of experimental device.

steel chamber of height 72 cm and 15 cm in diameter with a dust dropper, which is fitted inside the dust chamber.

To evacuate the chambers to a base pressure of 2×10^{-6} mbar, a diffusion pump (1000 l/s) backed by a rotary pump (600 l/min) is used. Hydrogen and argon gas are fed to the plasma chamber using two digital flow controllers (AALBORG make) attached to the plasma chamber. A full line cusped magnetic field confinement system is designed to improve the plasma confinement. It is a cylindrical shaped cage made up of stainless steel channels filled up with cube shaped strontium ferrite magnets having 1.2 kG surface field strength. Two numbers of Thoriated tungsten filaments (total length 150 mm and diameter 0.25 mm), arranged on a non-magnetic stand, are fitted on both the ends of the magnetic cage. Hydrogen plasma is produced by striking a discharge between incandescent tungsten filaments and grounded magnetic cage. Grounded magnetic cage serves as the anode. Ionizing electrons emitted from hot tungsten filaments are accelerated by applying 80 V discharge voltage.

The setup is also equipped with dust current measurement unit plasma parameter measurement system and plasma emission spectrometer system. Dust current measurement unit consists of a sensitive electrometer (Keithley Instruments, model: 6514) attached to a Faraday cup (see Fig. 2). The outer shell of the Faraday cup (FC) is electrically grounded. Apart from that, a strong transverse magnetic field is created by two permanent magnets of surface field strength 1.2 kG near FC aperture to prevent any lighter charged particles like electrons and ions to enter into the FC. The incoming charged dust particle strikes the copper plate inside the FC and flows to the ground through electrometer producing the dust current (~nA), which can be directly measured from electrometer reading.

The bandwidth of the electrometer is ~100 kHz as per the electrometer specification. Permanent magnets which are arranged on the both side of the FC significantly reduce stray current (background noise) and it is of the order of $\sim 10^{-12}$ A for different discharge conditions. It is ensured that the strength of the magnetic field does not affect the dust motion. So, the current recorded in the electrometer is due to the charged dust grain only.

Plasma parameters are measured by an advanced Langmuir probe system (Hiden Analytical), which consists of a cylindrical probe tip made of Tungsten of 0.15 mm in diameter and a length of 10 mm. The probe voltage is swept from −80 V to +80 V and the plasma parameters are measured in each 0.25 V division.

To estimate the relative contribution of hydrogen and argon ions at different percentage of argon flow rate in hydrogen plasma, a ½ m digikrom spectrometer (CVI Laser Corp, USA, Digikrom Model DK 480) is used. The system consists of a photo multiplier tube (PMT: model AD110, wavelength range: 185–930 nm) and a grating with 1200 grooves/mm. The system is used to measure the spectral line intensities within the region of $\lambda = (400–850)$ nm.

The dust density, measured by laser scattering technique,[28] is found to be of the order of $\sim 10^5\,\mathrm{cm}^{-3}$. For a uniform size dust grain, the charging time depends on plasma

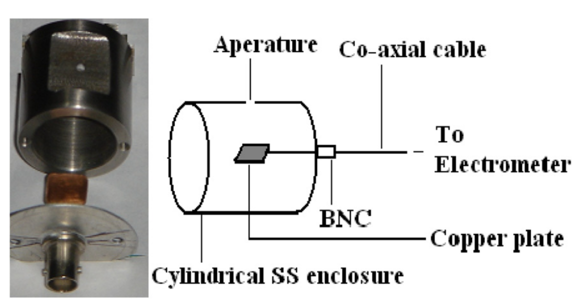

FIG. 2. Image and schematic of Faraday cup.

density, electron temperature, and ion mass.[29,30] The charging time of dust in plasma is given by[30]

$$\tau = K_\tau \frac{(kT_e)^{1/2}}{an}, \tag{2}$$

where K_τ is a function of T_i/T_e and m_i/m_e, a is the radius of the dust grain, and n is the plasma density. In the present experimental setup with available plasma parameters, the dust charging time inside the plasma chamber is ~few microseconds; whereas the time taken by a dust particle to cross the plasma volume under gravity is ~0.03 s. So, the dust grains have enough time to reach the equilibrium charge state within the plasma volume while passing through it.

III. RESULTS AND DISCUSSIONS

Keeping the hydrogen flow rate constant at ~8 sccm, different percentage of argon is added with hydrogen gas using two different digital flow controllers. For different percentage of argon flow rate, the total working pressure changes from 8×10^{-4} mbar to $\sim 1.8 \times 10^{-3}$ mbar.

A. Effect of argon addition on plasma parameters

Plasma parameters are measured at the center of the plasma column for different percentage of argon additive hydrogen plasma at discharge voltage of 80 V. Changing the filament current, the discharge current is varied from 100 to 700 mA. Plasma parameters and dust charging profile are examined at each 200 mA interval.

The limitations of the Langmuir probe measurements and the source of errors in probe measurement are explained by different researcher.[31–33] A large number of nonideal effects may create errors in the probe measurements. The imperfect designing of the probe tip on basis of the plasma parameters and the presence of magnetic field can yield significant error in the probe measurement. Other important factors which can influence the probe measurement are designing of the probe holder,[33,34] probe surface contamination,[35] change in work function of the probe tip, oscillation of the plasma potential,[36–38] and the background pressure (collisional effect).

In the present experiment, an advanced Langmuir probe system (Hiden analytical Ltd make) is used to measure the plasma parameters at the centre of the plasma column in the magnetic field free region. As the experiment is carried out in the pressure range of 10^{-3}–10^{-4} mbar and the observed plasma density is of the order of $\sim 10^{16}/\mathrm{m}^3$, thus, the influence of magnetic field and collisional effect on the probe measurement can be neglected. Godyak *et al.*[39] suggest that when the probe scanning time *(T)* is very less than the probe thermal equilibrium time (τ) (usually around 1 s), then probe temperature and probe work-function remain constant. For such condition, the *I-V* characteristics of the probe are unaffected by the probe work-function. Fast probe scanning times, in the range of milliseconds or less, avoid the convolution effect caused by a change in the probe temperature.[40] The probe scan time *(T)* of the Langmuir probe system used in our present experiment is very less than the probe thermal equilibrium time (τ) and before each scan, the probe is cleaned by applying a high positive voltage automatically. So, the errors which are often occurred in the probe measurement are considered as very less in the present measurement.

1. Effect on electron density

The *I–V* probe characteristic allows us to find the different plasma parameters like electron density, positive ion density, electron temperature, plasma potential, etc. When the probe biasing potential (V_b) is equal to the plasma potential (V_p), the *I–V* characteristic exhibits a "knee structure." For large positive probe biasing voltage $V_b > V_p$, the probe current is due to the collection of plasma electrons only which is known as the electron saturation current $I_{e\mathrm{sat}}$. The probe current at V_p is considered as electron saturation current which can be determined by intersection method and is used to estimate electron density[10] using the relation given below

$$n_e = \frac{I(V_p)}{A_p}\left(\frac{2\pi m_e}{e^3 \kappa T_e}\right)^{1/2}, \tag{3}$$

where A_p is the probe surface area, e is the elementary charge, m_e is the mass of the electron, T_e is the plasma temperature, and $I(V_p)$ is the electron saturation current at plasma potential (V_p). V_p is estimated from the zero of the second derivative of the *I-V* characteristics while T_e is determined from the slope of the straight line fitted to the transition regions of the *I-V* characteristics.

The variation of electron density at different percentage of argon additive hydrogen plasma is shown in Fig. 3. It is seen in Fig. 3 that as the percentage of argon flow rate increases, the electron density increases upto 30%. It is known that argon is easier than hydrogen to ionize (~3 times larger ionization cross section for argon atoms compared to that of hydrogen molecules for an electron energy of ~100 eV).[10,12] The shape of the electron energy distribution

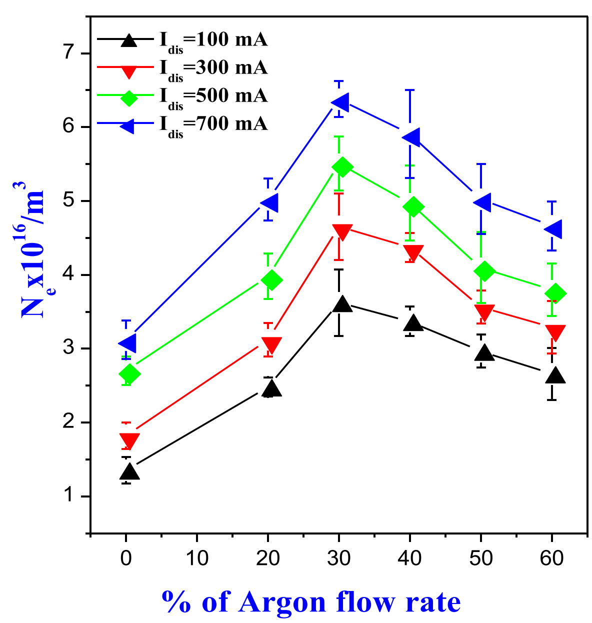

FIG. 3. Variation of electron density at different percentage argon flow rate.

function changes due to the addition of argon in hydrogen plasma, which can induce the change in the rate coefficients related to electron-induced inelastic collision processes.[41] Therefore, upto 30% addition of argon flow rate with hydrogen, an increase in electron density (by a factor ~2) is observed. Beyond 30% addition of argon flow rate, the electron density slightly decreases.

The different important inelastic processes between hydrogen species, argon species, and the electrons are described in details by Bogaerts and Gijbels.[13] The degree of ionization in the present experiment is very low (of the order of 10^{-3}), so the density of ionic H-species (H^+, $H_2{}^+$, $H_3{}^+$, H^-) are assumed to be negligible compared with the neutral atoms and molecules. Based on the rate coefficients, as described by Bogaerts and Gijbels,[13] the most dominant reactions in argon mixed hydrogen glow discharge are listed in Table I.

From the rate coefficient of the above reactions, it is found that reactions (1) and (3) are the most dominant reactions amongst them. The increase in argon flow rate increases the Ar^+ ions which in turn increases the ArH^+ ions in the hydrogen plasma background. Thus, at higher percentage of argon flow rate (above 30%), due to the reaction (1) followed by reaction (3), a slight decrease of electron density is observed (shown in Fig. 3). It is observed that for a clean hydrogen plasma, the electron density increases continuously from $1.35 \times 10^{16}\,\mathrm{m}^{-3}$ to $1.78 \times 10^{16}\,\mathrm{m}^{-3}$ for $I_{dis} = 100\,\mathrm{mA}$ when working pressure changes from 8×10^{-4} mbar to $\sim 2 \times 10^{-3}$ mbar. From Fig. 3, it is clear that addition of argon in hydrogen plasma enhances the electron density more significantly compared to the clean hydrogen plasma in the same working pressure range.

2. Effect on ion saturation current

Fig. 4 shows the variation of positive ion saturation current for different percentage of argon additive hydrogen plasma. As argon additive hydrogen plasma contains various positive ion species (Ar^+, ArH^+, H^+, $H_2{}^+$, $H_3{}^+$, etc.), it is essential to define ion density w. r. t effective ion mass or mean ion mass $(M_{i(X)})$. The ion current $I_i(V)$ collected into the cylindrical probe can be approximated as[42]

$$I_i(V) \approx \mathrm{e}A\Sigma n_{is(X)} \left(\frac{T_{eff}}{M_{i(X)}}\right)^{\frac{1}{2}} \left(1 + \frac{eV}{T_{eff}}\right)^{\frac{1}{2}}, \quad (4)$$

where $n_{is(X)}$ is the density of ion species X (X = Ar, H, H_2, H_3, and ArH) at the probe sheath edge, T_{eff} is the effective

TABLE I. List of important reactions in argon mixing hydrogen glow discharge. Reprinted with permission from A. Bogaerts and R. Gijbels, J. Anal. At. Spectrom. **15**, 441–449 (2000). Copyright 2000 Royal Society of Chemistry Publishing.[13]

Sl. No	Reactions	Rate co-efficient
1.	$Ar^+ + H_2 \rightarrow ArH^+ + H$	$\sim 10^{-9}\,\mathrm{cm^3\,s^{-1}}$
2.	$ArH^+ + H_2 \rightarrow Ar(fast) + H_3{}^+$	$\sim 3\text{–}5 \times 10^{-9}\,\mathrm{cm^3\,s^{-1}}$
3.	$e^- + ArH^+ \rightarrow Ar + H^*$	$\sim 10^{-7}\,\mathrm{cm^3\,s^{-1}}$
4.	$Ar_m{}^* + H_2 \rightarrow Ar + H_2$ or $H_2{}^+$	$\sim 10^{-10}\,\mathrm{cm^3\,s^{-1}}$
5.	$Ar_m{}^* + H \rightarrow ArH^* \rightarrow Ar_m + H^*$	$\sim 2 \times 10^{-10}\,\mathrm{cm^3\,s^{-1}}$

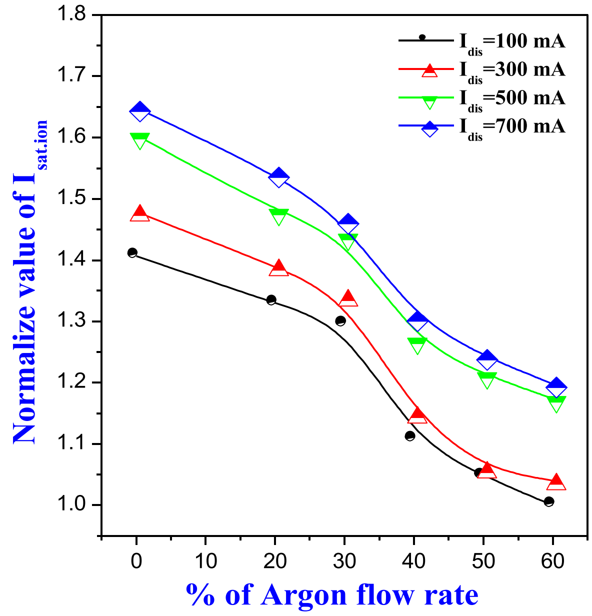

FIG. 4. Variation of positive ion saturation current at different percentage of argon flow rate.

electron temperature, and $M_{i(X)}$ is the effective ion mass or mean ion mass. Due to the presence of different positive ion species, the effective ion mass in such plasma changes. Substituting the ion saturation current $I_i(V)$, electron temperature (kT_e), and ion density (N_i) values, the effective ion mass can be calculated[43] using Eq. (4).

From Fig. 4, it is seen that the positive ion saturation current decreases with the addition of argon into the hydrogen plasma. The most probable factors which are responsible for decrease of positive ion saturation current measured by Langmuir probe, are (a) due to the increase of effective ion mass (shown in Fig. 5), (b) at higher percentage (above

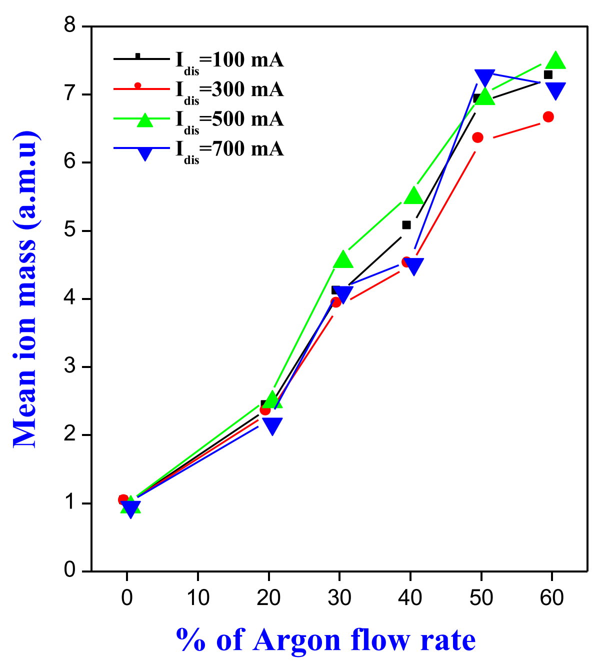

FIG. 5. Variation of mean ion mass at different percentage of argon flow rate.

30%) of argon flow rate, Ar^+ ion loss predominantly through the H atom transfer (due to reaction 1), and (c) due to the decrease of degree of dissociation of hydrogen molecule. The variation of effective ion mass for different percentage of argon flow rate is shown in Fig 5. As Ar^+ ion is heavier than H^+ ion, so, an increase in argon flow rate in hydrogen plasma increases the mean ion mass continuously.

3. Effect on electron temperature

The electron temperature is calculated from the transition region of the *I-V* characteristics of the Langmuir probe. The determination of electron temperature is facilitated by re plotting the probe current on a semi-log scale.[44] The slope of the exponentially decreasing portion on the semi-log plot gives electron temperature (*Te*). Fig. 6 shows the changes in electron temperature at different percentage of argon flow rate. There is a significant change in electron temperature with the increase of argon percentage. Due to the larger cross-section of argon compared to hydrogen, the collisions between electron and argon species increase. As a result, there is a drop of electron temperature observed (drops by a factor of ~0.5 for different discharge current) till 40% addition of argon flow rate with hydrogen plasma. Beyond 40% value, the electron temperature becomes constant. For clean hydrogen plasma, the change in electron temperature is almost negligible compared to the argon mixed hydrogen plasma within the pressure regimes 8×10^{-4} mbar to 2×10^{-3} mbar in our experimental device.

4. Effect on H^- ion density

Optical emission spectroscopy (OES) measurement is carried out to study the effect on H^- ion production and degree of dissociation due to the addition of argon. Different researchers have reported earlier that addition of argon into the hydrogen discharge enhances the H^- ion density.[4–7] Fantz and Wünderlich[45] reported that H-Balmer line intensity ratio (H_α/H_β) can be used to estimate H^- ion density.

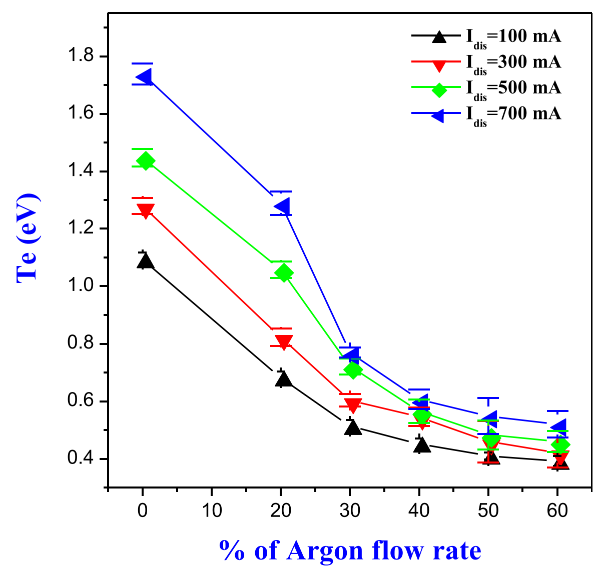

FIG. 6. Variation of T_e at different percentage of argon flow rate.

They reported that the line intensity of H_α/H_β is very sensitive to H^- ion density. The line intensity of H_α/H_β increases with the increase of H^- ion density. Here, line intensity of H_α/H_β is observed to identify the negative ion density due to the addition of argon into the hydrogen discharge. The line intensity of H_α/H_β is shown in Fig. 7. From the graph, it is seen that the variation of line intensity of H_α/H_β is not prominent with the addition of argon in hydrogen plasma. It implies that the production of H^- ion density due to the argon addition is insignificant in the present experiment. Therefore, the effect of H^- ion density on plasma parameters as well as on dust charging is neglected.

5. Effect on degree of dissociation

The line intensity of H_α and H_β for different percentage of argon flow rate is shown in Fig. 8. The intensities of these lines contain information about the population densities of excited levels of atomic hydrogen.[42]

The line intensities I_{656}, I_{486}, and I_{750} are approximately given as

$$I_{656} \approx C^1{}_{i,j(H)} h\nu_{i,j(H)} \frac{A_{i,j(H)}}{\sum A_{i,j(H)}} n_e (k_{exc,H} n_H + k_{dis-exc,H_2} n_{H_2}), \tag{5}$$

$$I_{486} \approx C^2{}_{i,j(H)} h\nu_{i,j(H)} \frac{A_{i,j(H)}}{\sum A_{i,j(H)}} n_e (k_{exc,H} n_H + k_{dis-exc,H_2} n_{H_2}), \tag{6}$$

$$I_{750} \approx C_{i,j(Ar)} h\nu_{i,j(Ar)} n_e k_{exc,Ar} n_{Ar}, \tag{7}$$

where $k_{\mathrm{exc,H}}$ and $k_{\mathrm{exc,Ar}}$ are the excitation rate coefficients of the ground-state hydrogen atom and argon, respectively, $k_{dis-exc,H_2}$ is the dissociative-excitation rate coefficient of the ground-state molecular hydrogen, n_e is the electron density, n_X is the density of the neutral species X (=H, H_2, and Ar),

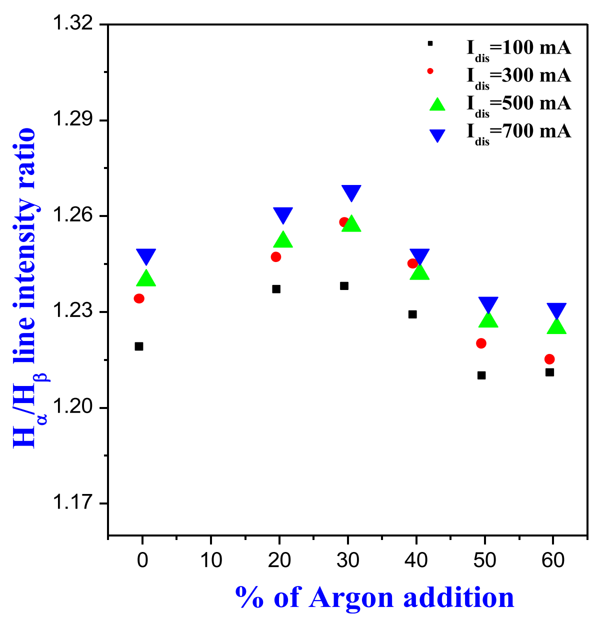

FIG. 7. Line intensity ratio of H_α/H_β at different percentage of argon flow rate.

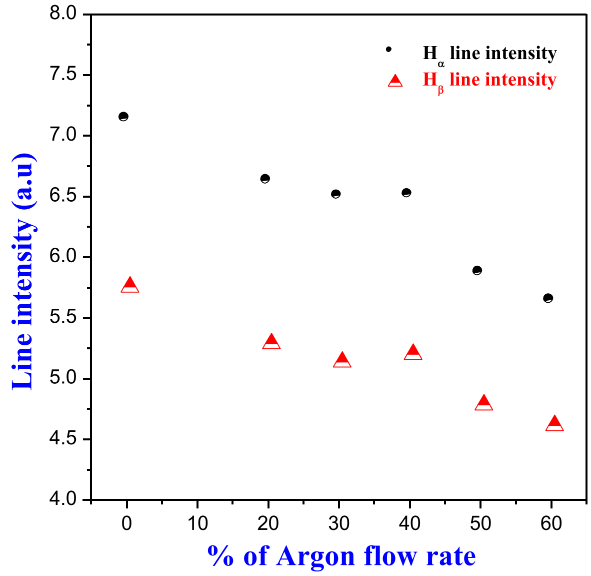

FIG. 8. Line intensity of H_α and H_β for different percentage of argon flow rate.

$C^1_{i,j(H)}$ and $C^2_{i,j(H)}$ are the constant related to the solid angle and spectral response of the optical system, h is the Planck's constant, $\nu_{i,j(X)}$ is the frequency of the transition from the i-level to the j-level, and $A_{i,j(X)}$ is the Einstein coefficient for the observed transition.

In low temperature laboratory plasma, the density of atomic hydrogen is almost negligible compared to the density of hydrogen molecules.[42] The dissociative excitation process of molecular hydrogen is a very important reaction in low temperature plasma. Thus, it is considered that the contribution of the dissociative excitation process of molecular hydrogen in the formation of H_α and H_β line intensities is dominant than the H atom excitation. Some important processes occurring in low pressure argon mixed hydrogen plasma are listed in Table II.

From the Fig. 8, it is seen that the line intensity of H_α and H_β decrease with the increase of argon flow rate. It indicates that the addition of argon in hydrogen plasma decreases the dissociative excitation of H_2 which in turn decreases the degree of dissociation.

For verification, the line intensity ratio of H_β/Ar-750.4 is measured for different percentage of argon flow rate (shown in Fig. 9). Different researchers[42,46–49] have reported that the line intensity ratios of I_H/I_{Ar} can be used to determine the degree of dissociation in hydrogen plasma using the following relation:

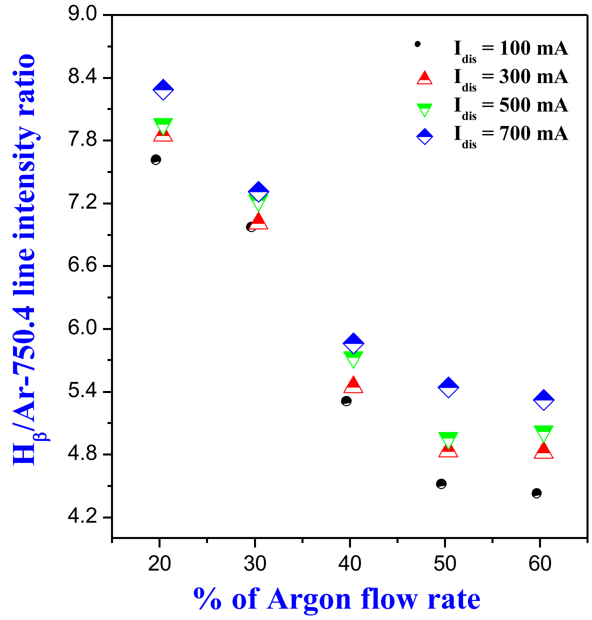

FIG. 9. Line intensity ratio of H_β/Ar-750.4 nm for different percentage of argon flow rate.

$$N_H = K\frac{I_H}{I_{Ar}}N_{Ar} \quad \text{with} \quad K = \frac{\alpha_{Ar}A_{Ar}\tau_{Ar}\lambda_{nn/}}{\alpha_H A_{nn/}\tau_{nn/}\lambda_{Ar}}, \tag{8}$$

where $\alpha_{Ar,H}$ are the rate coefficients for direct excitation of the corresponding Ar and H atom levels, $A_{Ar,\ nn^/}$ and τ are, respectively, the radiation transition probabilities and life times of the corresponding excited states and λ is the wavelength.

To determine the degree of dissociation, the selected combinations of line intensity ratios of (I_H/I_{Ar}) are I(H_α)/I(Ar 811), (H_β)/I(Ar750.4), and (H_γ)/(Ar750.4).[47] In the present work, we have chosen H_β and Ar-750.4 lines to estimate the degree of dissociation because the excitation cross sections of the hydrogen atoms and Ar have similar energy dependence.[46–49] It is seen from Fig. 9 that the line intensity ratio of

TABLE II. Some important processes occurring in H_2/Ar low-pressure plasmas.

Sl. No	Processes		E_{th} (eV)	Rate co-efficient at 3 eV ($m^3\ s^{-1}$)	Reference
1.	H atom excitation	H(n = 1) + e→H(n = 3) + e	12.09	k_H^{dir} 41.4	16
		H(n = 1) + e→H(n = 4) + e	12.75	8.86	
2.	H_2 dissociative excitation	H_2 + e→ H(n = 3) + H(n = 1) + e	16.57	k_H^{dis} 0.53	16
		H_2 + e→ H(n = 4) + H(n = 1) + e	17.23	0.01	
3.	Ar direct excitation	Ar(3p) + e→Ar(4p/[1/2]0) + e	13.48	k_{Ar}^{dir} 10.7	17
		Ar(3p) + e→Ar(4p[5/2]3) + e	13.08	11.2	
4.	Radiative de-excitation	H(n = 3)→H(n = 2) + hν(H_α-656 nm)	44.1	99.8	18
		H(n = 4)→H(n = 2) + hν(H_β-486 nm)	8.4	30.2	
		Ar(4p/[1/2]0)→Ar(4s) + hν(750 nm)	44.5		18
		Ar(4p[1/2]0)→Ar(4s) + hν(811 nm)	33.1		

TABLE III. List of important reactions in pure hydrogen plasma.

Sl. No	Reactions	Rate co-efficient ($m^3\ s^{-1}$)	Reference
1.	$e + H_2 \rightarrow H + H + e$	$k_1 = 1.2 \times 10^{-14}\ \exp(-10/T_e)$	50
2.	$e + H_2 \rightarrow H + H(2s) + e$	$k_2 = 4.71 \times 10^{-15}\ \exp(-15.9/T_e)$	50
3.	$e + H_2 \rightarrow H(2p) + H(2s) + e$	$k_3 = 1.78 \times 10^{-15}\ \exp(-28.34/T_e)$	50
4.	$e + H_2 \rightarrow H + H(n = 3) + e$	$k_4 = 3.74 \times 10^{-16}\ \exp(-18.0/T_e)$	50
5.	$e + H_2 \rightarrow {H_2}^+ + 2e$	$k_5 = 3.11 \times 10^{-14}\ \exp(-18.9/T_e)$	50 and 51
6.	$e + H_2 \rightarrow H + H^+$	$k_6 = 3.07 \times 10^{-16}\ \exp(-17.5/T_e)$	50
7.	$e + {H_2}^+ \rightarrow H + H(n \geq 2)$	$k_7 = 8.0 \times 10^{-14}\ \exp(-0.2T_e)$	50
8.	$e + {H_2}^+ \rightarrow H^+ + H + e$	$k_8 = 1.45 \times 10^{-13} \exp(-1.97/T_e)$	50
9.	$e + {H_3}^+ \rightarrow H_2 + H$	$k_9 = 1.55 \times 10^{-13}(300/T_g)$	52

H_β/I(Ar750.4) decreases as the percentage of argon flow rate increases. It is an indication about the decrease of degree of dissociation of hydrogen plasma as a result of argon addition, due to reduction in electron temperature.

Some important reactions for dissociation of molecular hydrogen species in hydrogen plasma are given in Table III.

In pure hydrogen plasma, hydrogen atoms are produced basically in the electron impact dissociation of H_2 (reaction 1 of Table III) and are lost in wall collisions, where they recombine to form H_2 ($H + \text{wall} \rightarrow H_2$). The density of molecular hydrogen ions (like ${H_2}^+$ and ${H_3}^+$) are very less compared to H_2, so the electron impact dissociation of molecular hydrogen ions can be neglected. Due to the decrease of electron temperature, the reaction rate of the above reaction decreases, which in turn reduces the degree of dissociation of H_2 with the increase of argon flow rate.

In H_2 mixed argon plasma, another process which can lead to the formation of H atoms is the dissociative excitation of H_2 molecules by Ar_m* atoms.[53] Ar_m* denotes exited Ar atom in metastable state. As the density of Ar_m* atoms is very less compared to H_2 and the production of Ar_m* atoms decreases continuously with reduction of electron temperature, so the quenching dissociation is not significant in our present work.

B. Effect on dust surface potential

The charge accumulated on the micron or sub-micron sized dust particles in plasma environment is one of the most important parameters in complex plasma, plasmas in semi-conductor or artificial diamond industries,[54] astrophysical plasma, fusion plasma,[55,56] etc.

The dust grain surface acquires a floating potential (ϕ_d) at which the electron and ion currents become equal in magnitude, i.e., $I_e + I_i = 0$. Substituting the electron (I_e) and ion (I_i) current, the floating potential of negatively charged dust grain can be estimated from the relation

$$-en_e\pi a^2\sqrt{\frac{8k_BT_e}{\pi m_e}}e^{\frac{e\phi_d}{k_BT_e}} + en_i\pi a^2\sqrt{\frac{8k_BT_i}{\pi m_i}}\left(1-\frac{e\phi_d}{k_BT_i}\right) = 0,$$

$$\Rightarrow \left(1-\frac{e\phi_d}{k_BT_i}\right) = \frac{n_e}{n_i}\cdot\sqrt{\frac{T_e}{T_i}\cdot\frac{m_i}{m_e}}.\exp\left(\frac{e\phi_d}{k_BT_e}\right), \tag{9}$$

where e, ϕ_d, k_B, a, Z_i, $n_{e\ (i)}$, $T_{e\ (i)}$, and $m_{e\ (i)}$ denote the electronic charge, dust grain surface potential, Boltzmann's constant, radius of dust grain, ionic charge, electron (ion) density, electron (ion) temperature, and mass of electron (ion), respectively.

The effect on dust surface potential for different effective ion mass and electron temperature are shown in Fig 10. The dust surface potential is calculated numerically for different effective ion mass and electron temperature by solving the Eq. (9). The black color line indicates the dust surface potential for different effective ion mass at constant electron temperature (1 eV), and the red color line indicates the dust surface potential for different electron temperature at constant effective ion mass (1 a.m.u). From Fig. 10, it is observed that when the electron temperature changes from 0.5 eV to 2 eV, the surface potential changes from −0.85 V to −2.6 V for constant ion mass 1 a.m.u. On the other hand, when the positive ion mass changes from 1 a.m.u to 8 a.m.u, the surface potential changes from −1.5 V to −2.2 V for constant electron temperature 1 eV. It indicates that the effect of electron temperature on dust surface potential is prominent compared to that of ion mass.

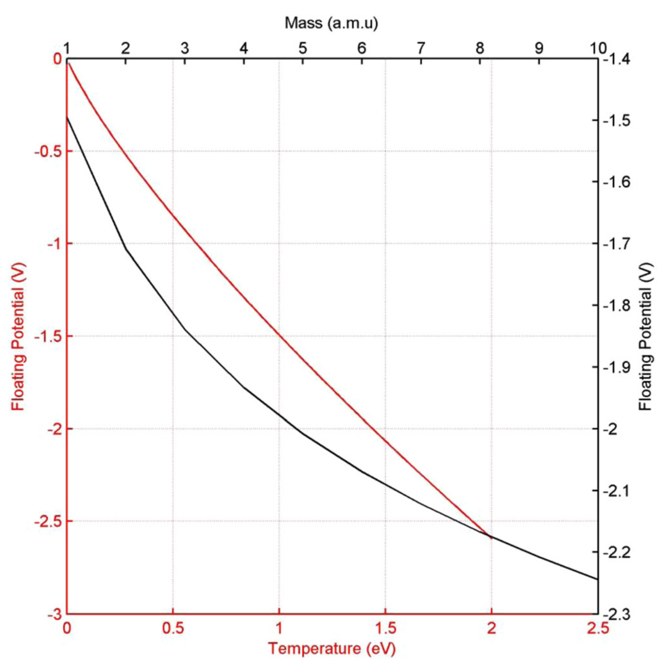

FIG. 10. Variation of dust surface or floating potential (a) at different electron temperature for a constant positive ion mass (1 a. m. u) [Red line and axes] and (b) at different positive ion mass for a constant electron temperature (1 eV) [Black line and axes].

C. Effect on dust charging

The dust grain will become charged negatively till the dust grain surface acquires a floating potential at which the electron and ion currents become equal in magnitude. Due to the accumulation of charges, the spherical dust particle behaves as a spherical capacitor. Therefore using the capacitance model,[57] the charge on the spherical dust grains can be calculated by using the relation

$$q_C = eN_C = 4\pi\varepsilon_0 a\phi_d, \tag{10}$$

where q_c is the total charge on grains, e is the electronic charge, N_C is the number of charges accumulated on the dust grain, a is the size of the dust grain, and ϕ_d is the floating potential of the dust grain relative to the plasma potential.

The charge accumulated on tungsten dust particles (calculated using capacitance model) at different percentage of argon addition into the hydrogen plasma is shown in Fig. 11. In our present experiment, it is found that the dust charge changes from $4.56 \times 10^3 - 2.43 \times 10^3$ times of electronic charge for different percentage argon flow rate which is analogous with the results observed by Kersten *et al.*[58] and Melzer *et al.*[59] for one component positive ion plasma. It is seen that the charge accumulated on dust particle decreases with the addition of argon into the hydrogen plasma.

Confirming the changes in charge accumulation, the combination of Faraday cup and electrometer are used as an additional technique to observe the dust current profile. The dust current for different percentage of argon flow rate in hydrogen plasma is shown in Fig. 12. As the experiment is carried out in a low pressure multi-cusp dusty plasma device, the different discharging processes[60–62] are negligible. Thus, it is assumed that the dust current is proportional to the charge carried by the dust grains.

From the Fig. 11 and 12, it is seen that the dust charge and dust current decreases up to 40% addition of argon flow

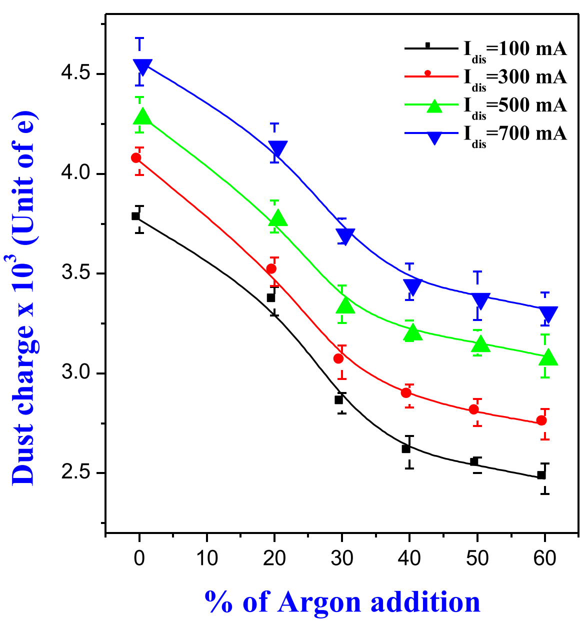

FIG. 11. Dust charge (using capacitance model) at different percentage of argon flow rate.

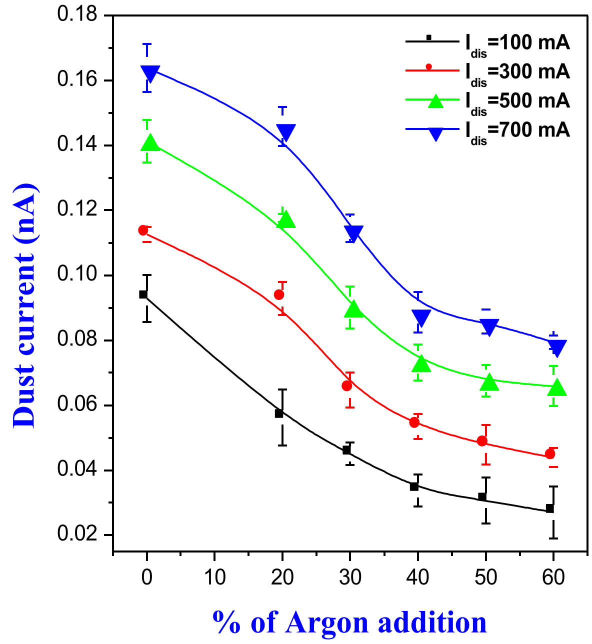

FIG. 12. Dust current at different percentage of argon flow rate.

rate in hydrogen plasma. But beyond 40% addition of argon flow rate, the dust current and dust charge get saturated. It is observed in our earlier work that the charge accumulated on the dust grain in plasma directly depends on the electron temperature rather than plasma density.[26,27]

The charge accumulated on dust grains is directly proportional to the dust surface potential[57] ($Q \propto \phi_d$). Since the influence of positive ion mass on dust surface potential (ϕ_d) is insignificant compared to electron temperature (shown in Fig 11), so the influence of effective positive ion mass on dust charge is not observed. The mass of the positive ions changes the charging time only. Since the dust grains have enough time to reach the equilibrium charge state within the plasma well, the effect of positive ion mass is not seen in the present case.

Comparing the results shown in Figs. 11 and 12 with electron density, positive ion density, positive ion mass, electron temperature, and degree of dissociation, it is observed that the dust charge and dust current follows the trend of the electron temperature only. In our recent publication,[63] it is reported that the effect of working pressure on dust charging for clean hydrogen plasma is almost negligible within the pressure regimes 8×10^{-4} mbar to 2×10^{-3} mbar. From the above results, a prominent effect on dust charging due to the addition of argon in hydrogen plasma is observed.

IV. CONCLUSION

From the present experiment, it is found that the electron density increases by a factor ~2 and the electron temperature reduces by a factor 0.5 till addition of argon flow rate with hydrogen gas goes upto 30%. Beyond 30% addition of argon with hydrogen flow rate, the electron density start decreasing; whereas, the electron temperature becomes constant for 40%–60% addition of argon flow rate.

From the OES study, it is found that due to argon addition, the degree of dissociation of hydrogen plasma decreases due to lowering of electron temperature. The atomic hydrogen concentration in the ion source is a crucial plasma parameter for efficient production of negative hydrogen ion because one of the main loss mechanisms of the negative ions is via collisions with atoms.[47] Again, the increase in low-energy electron density ($Te \leq 1\ eV$) enhances the H^- ion density in volume assisted negative ion source.[64] Thus, 30% addition of argon flow rate in hydrogen discharge may be beneficial for efficient H^- ion production in volume assisted negative hydrogen ion source.

From the dust charging profile (Figs. 11 and 12), it is observed that the dust current and dust charge decrease significantly till addition of argon flow rate in hydrogen plasma is around 40%. Beyond 40% of argon flow rate, the reduction of dust current and dust charge are insignificant. Comparing the dust charging results, the variation of electron density (in Fig. 3) and dust charging profile (in Figs. 11 and 12) for different percentage of argon addition, opposite trend is observed. On the other hand, it is seen that both dust charging profile (in Figs. 11 and 12) and the electron temperature (in Fig. 6) follows similar trend. Like electron temperature, dust charging drops till 40% addition of argon with hydrogen plasma and it gets saturated beyond 40%. It indicates the strong dependence of electron temperature on dust charging in low pressure plasma. Again, comparing the dust charging results (Figs. 11 and 12) with Fig. 5, it is seen that the dust charging gets saturated beyond 40% of argon addition, although the effective ion mass increases continuously with the argon addition. Thus, based on above observations, it can be concluded that electron temperature plays the dominant role on dust charging. The effect of plasma density, degree of dissociation and positive ion mass on the dust charging is insignificant. From the present experiment, it is found that addition of argon into the hydrogen plasma can be used as a tool to control the dust charging in low-pressure hydrogen plasma.

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